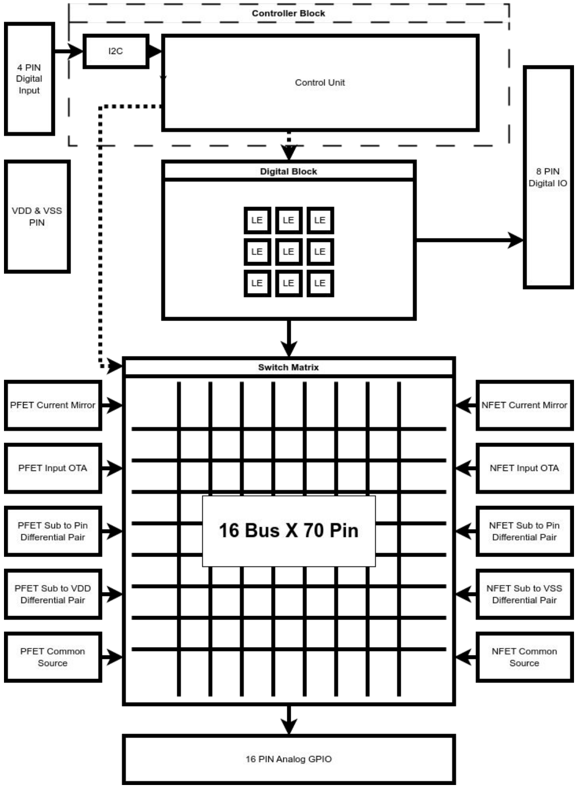
M6_Low Noise Anomaly



Mixed-Signal MOSbius Style:

4x Programmable Digital CLB
1x P-OTA
1x N-OTA
4x PFET Current Mirror
4x NFET Current Mirror
8x Individual PFET
8x Individual NFET

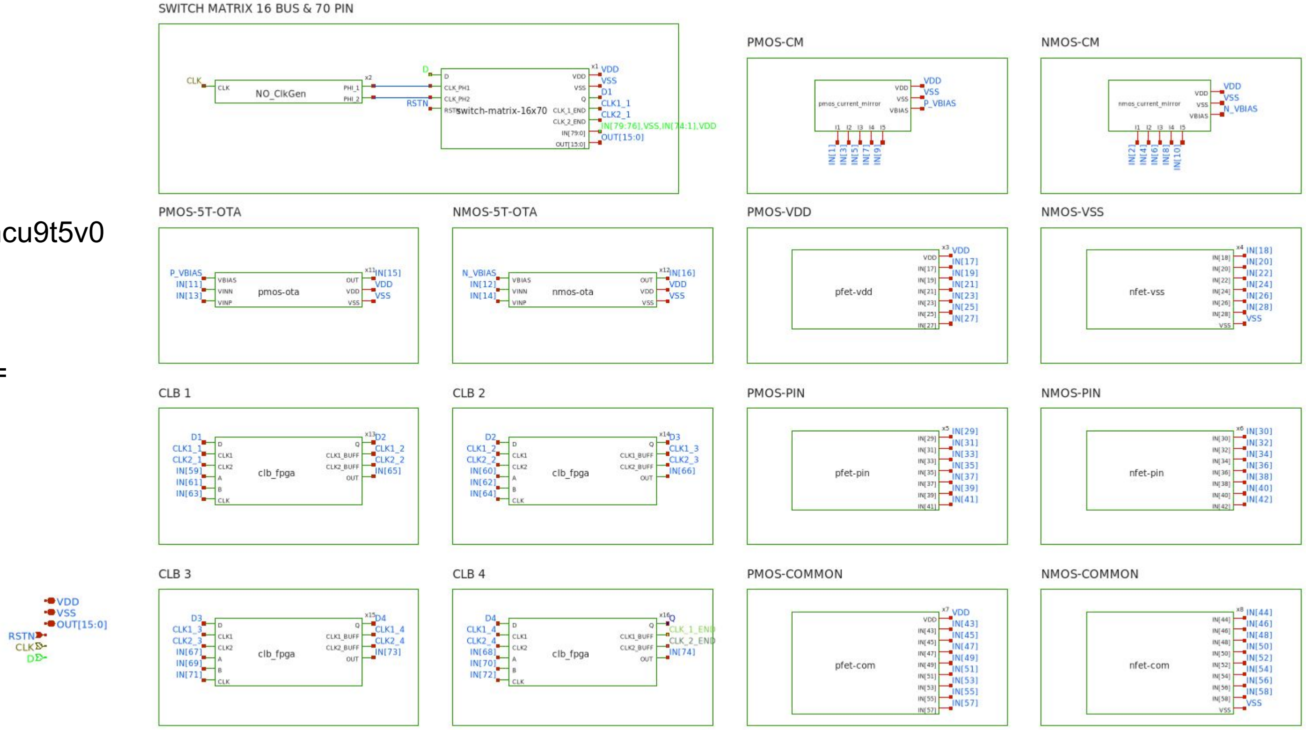
M6_Low Noise Anomaly

Top-Level Module

std-cell =
gf180mcu_fd_sc_mcu9t5v0

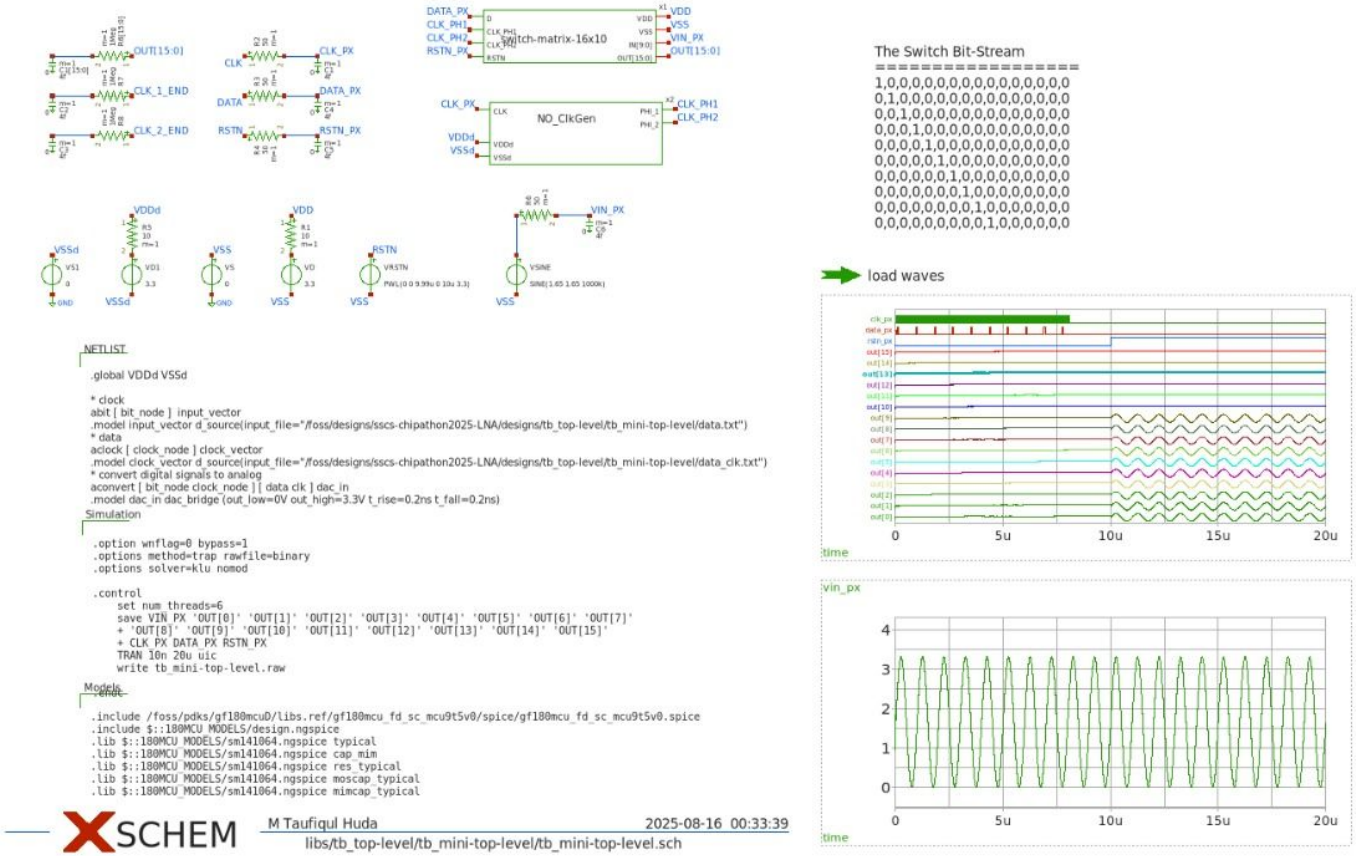
VDD = 3v3

Switch-matrix size =
75 Pin X 16 Bus

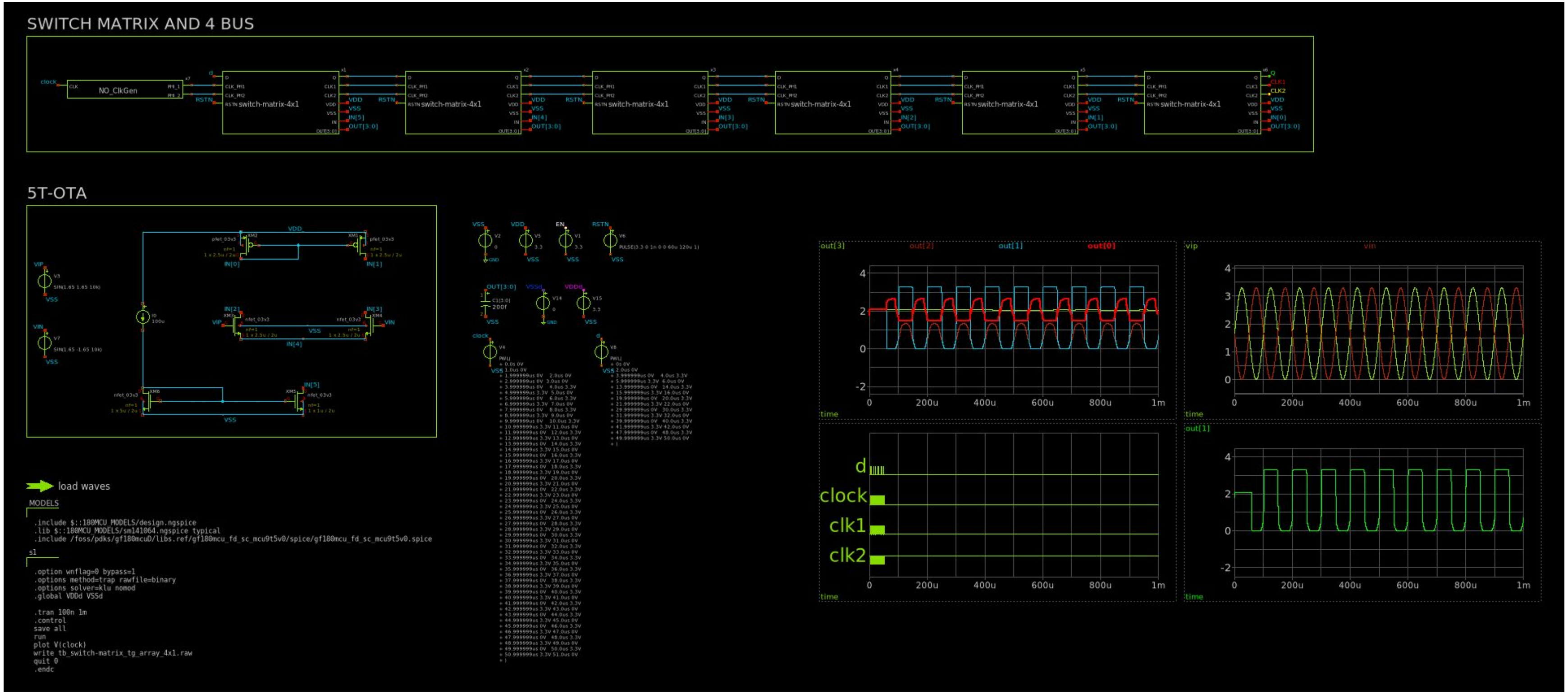


Mini TB top-level Simulations

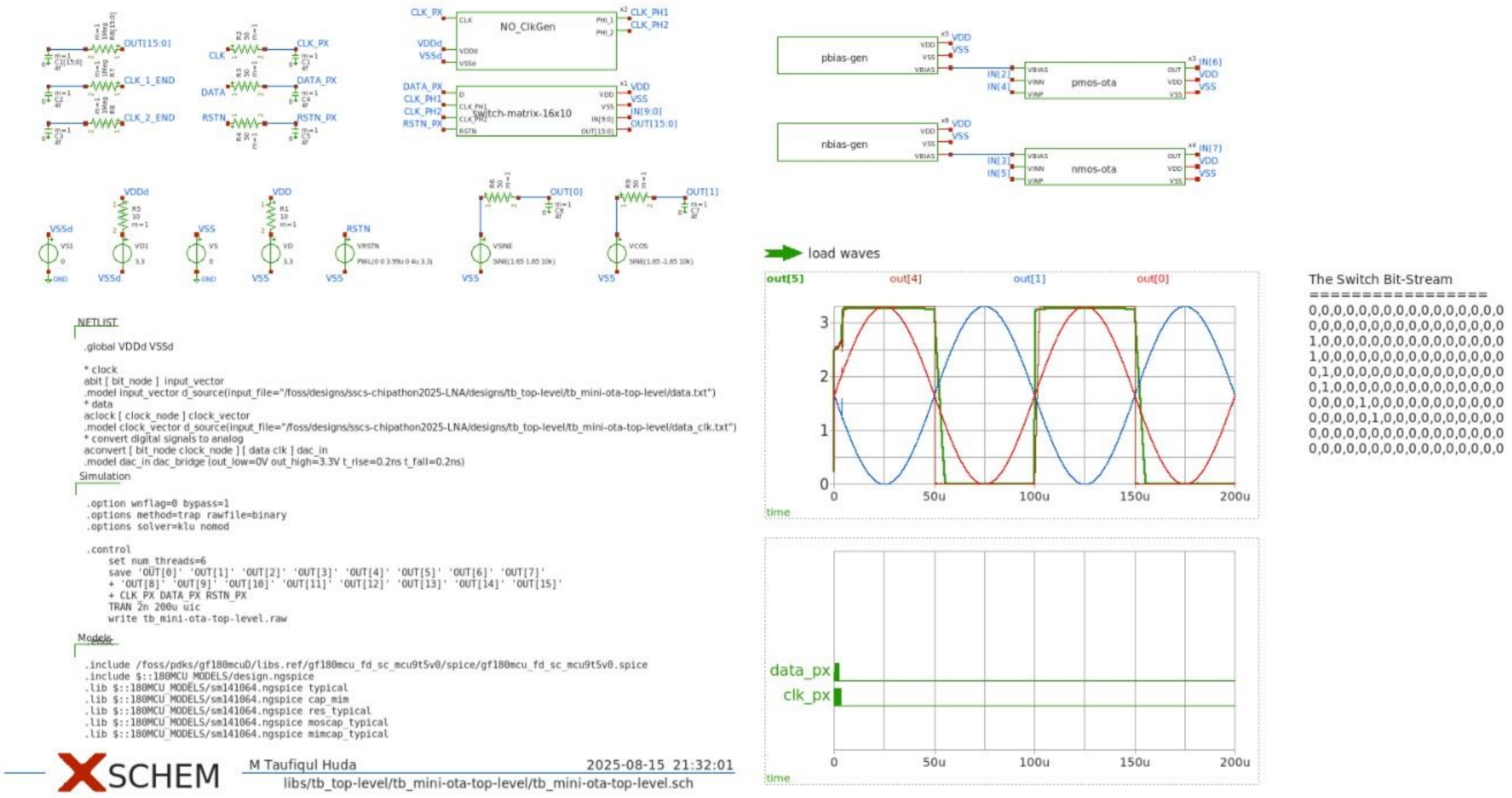
M6_Low Noise Anomaly : Connection test W/ MOSbius Architecture



M6_Low Noise Anomaly : Single FET test W/ MOSbius Architecture



M6_Low Noise Anomaly : OTA test W/ MOSbius Architecture



M6_Low Noise Anomaly : CLB (Digital) test W/ MOSbius Architecture

load waves

Models

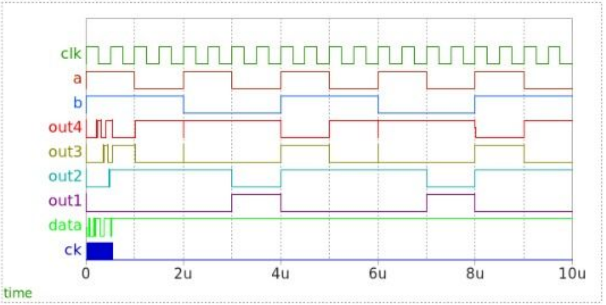
```
.include /foss/pdks/gf180mcuD/libs.ref/gf180mcu_fd_sc_mcu9t5v0/spice/gf180mcu_fd_sc_mcu9t5v0.spice
.include $::180MCU_MODELS/design.ngspice
.lib $::180MCU_MODELS/sm141064.ngspice typical
```

Simulation

```
.global VDDd VSSd
* clock
abit [ bit_node ]  input_vector
.model input_vector d_source(input_file="/foss/designs/sscs-chipathon2025-LNA/designs/libs/tb_fpga/tb_clb/data.txt")
* data
aclock [ clock_node ] clock_vector
.model clock_vector d_source(input_file="/foss/designs/sscs-chipathon2025-LNA/designs/libs/tb_fpga/tb_clb/data_clk.txt")
* convert digital signals to analog
aconvert [ bit_node clock_node ] [ data ck ] dac_in
.model dac_in dac_bridge (out_low=0V out_high=3.3V t_rise=0.2ns t_fall=0.2ns)

.control

    save all
    TRAN 0.2n 10u
    write tb_clb.raw

.endc
```

CLB Register
==
1000,[0/1] => NAND
0111,[0/1] => OR
0001,[0/1] => NOR
1110,[0/1] => AND
1001,[0/1] => XNOR
0110,[0/1] => XOR

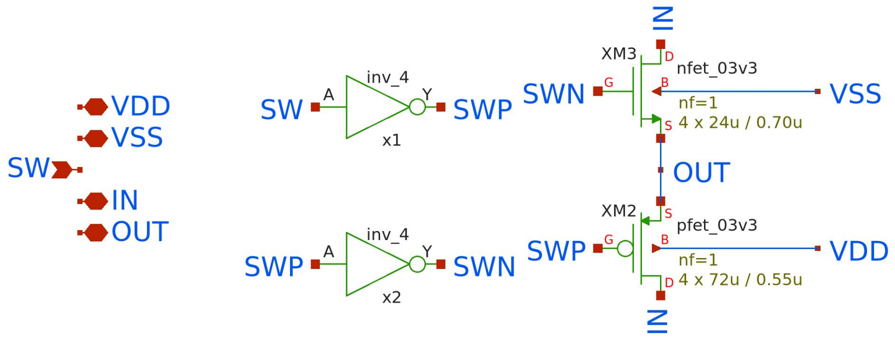
NB : [0/1] is MSB stand for [0/1] => [Sequential(FF)/Combinational(ABC)]

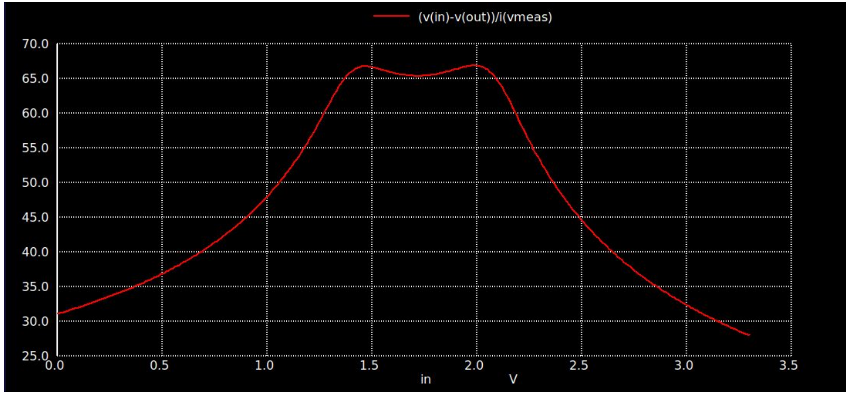


Spec and Test Every Module

M6_Low Noise Anomaly

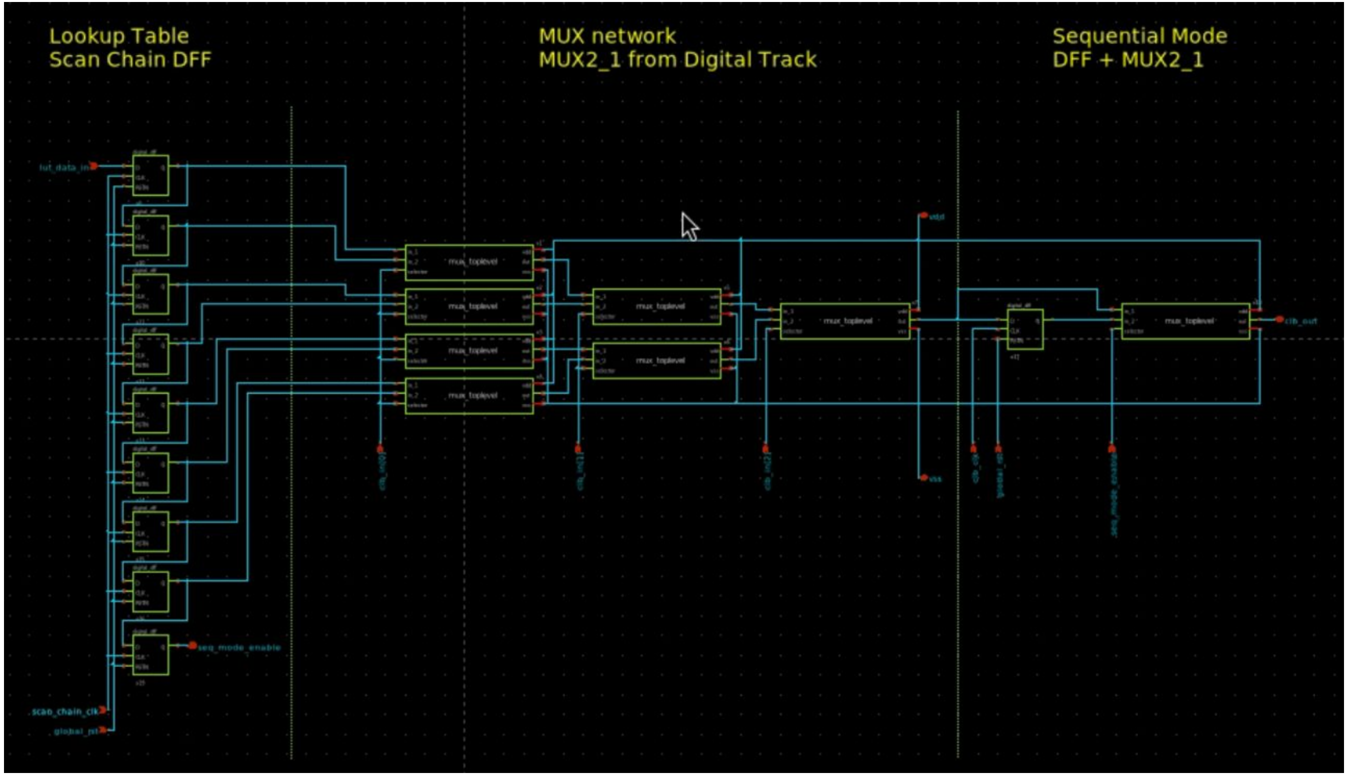
Switch-matrix Transmission Gate Specs



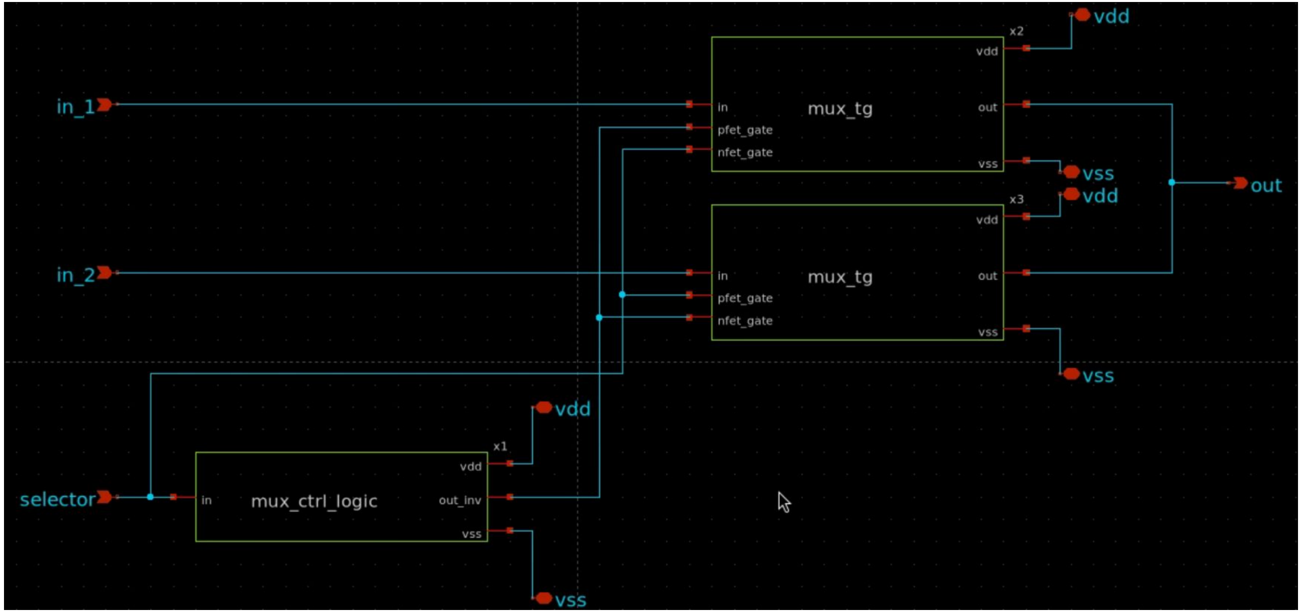


Already achieve < 100 Ohm each Switch Cell,
[Spec is MET]

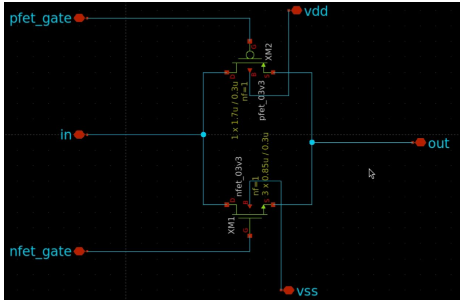
M6_Low Noise Anomaly : CLB Schematic



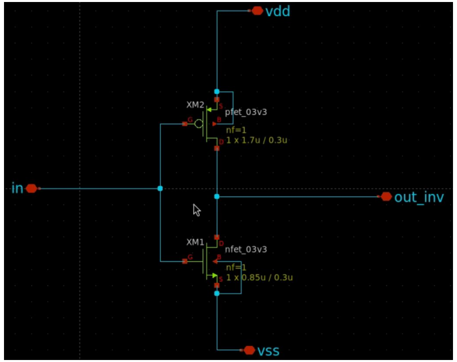
M6_Low Noise Anomaly : CLB Internals – MUX2_1



Top Level Schematic for MUX2_1

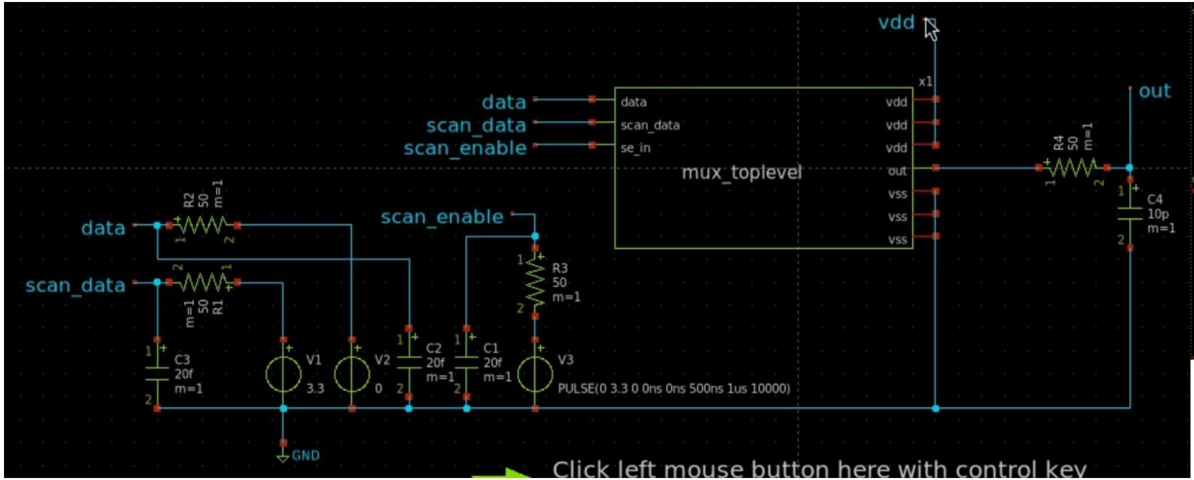


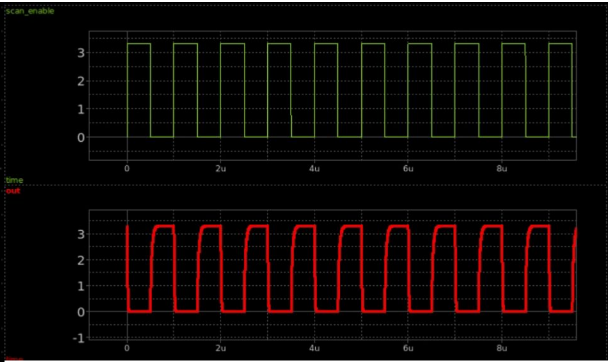
Transmission Gate



Inverter

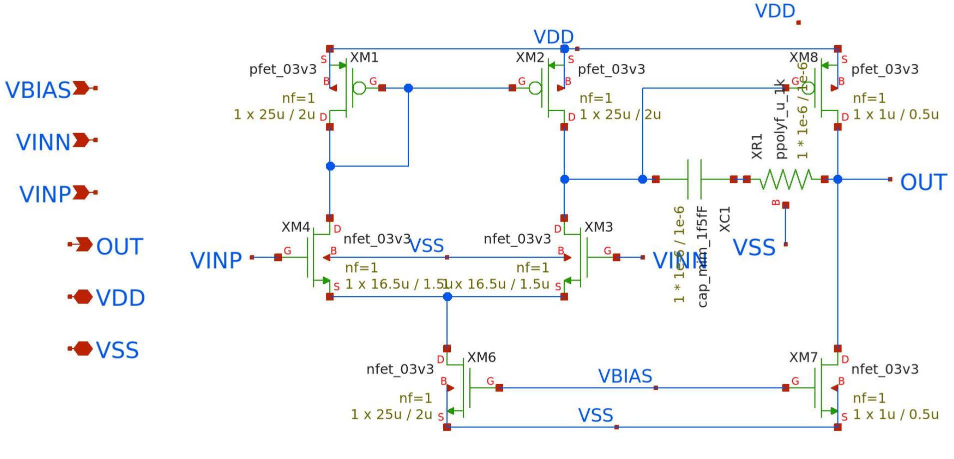
M6_Low Noise Anomaly : CLB Internals – MUX2_1

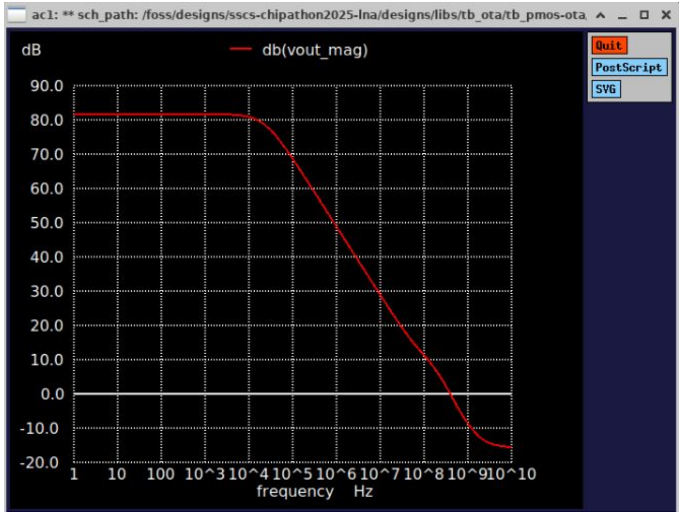




Testbench schematic

M6_Low Noise Anomaly

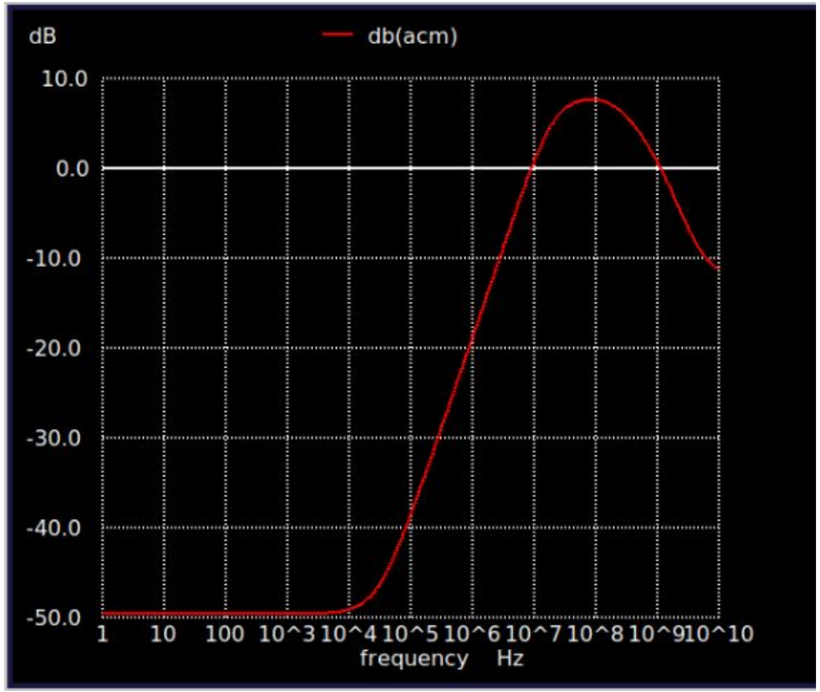




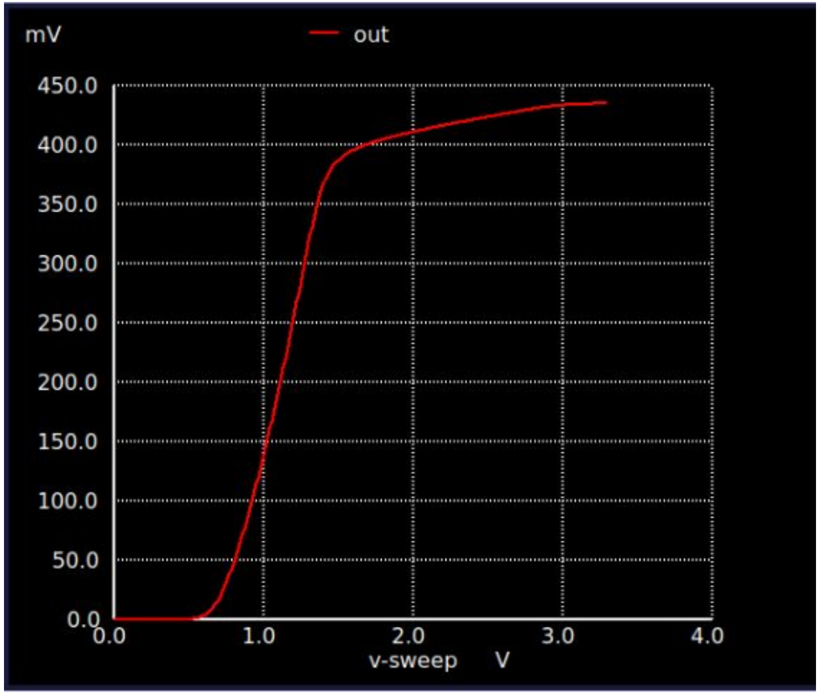
Operating Points	
Corner Case	Typical
Ibias	10uA
VDD	3.3 Volts

2-stage nmos OTA

M6_Low Noise Anomaly

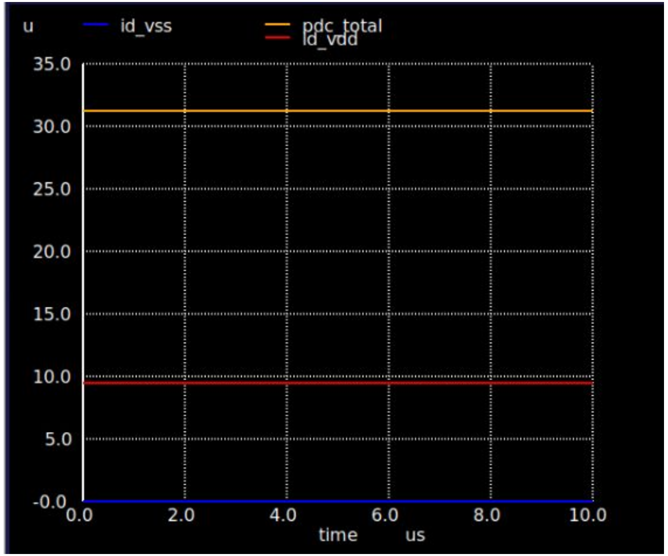


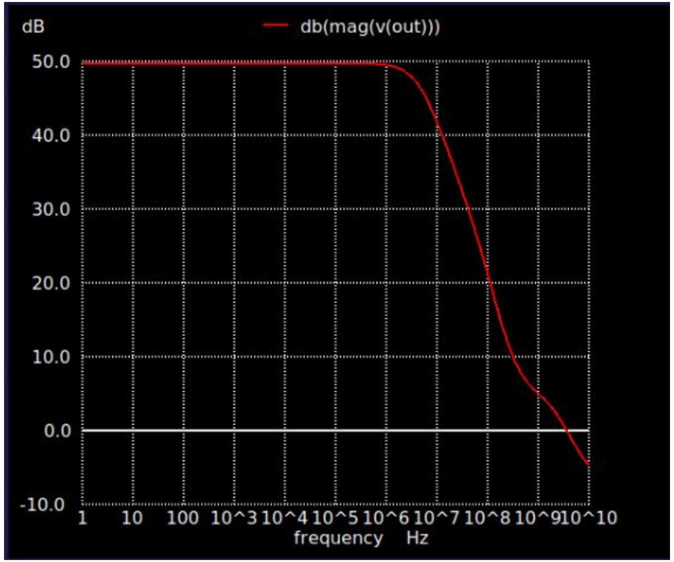
Common-Mode Gain



Input Common Mode Range

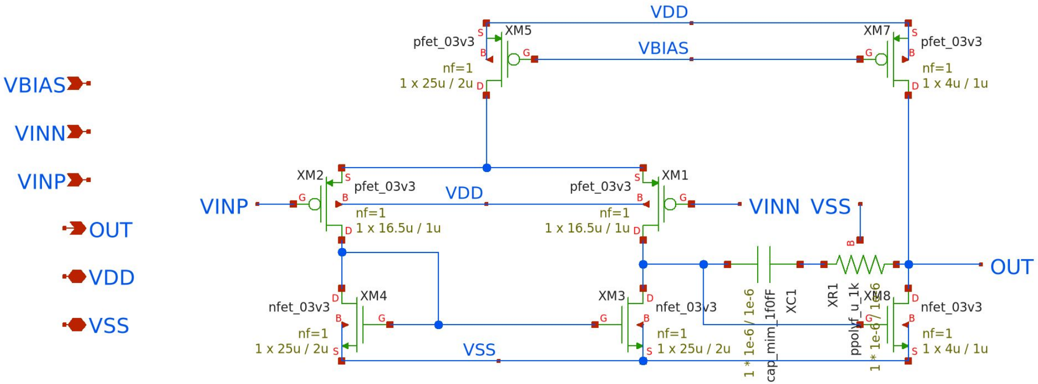
M6_Low Noise Anomaly

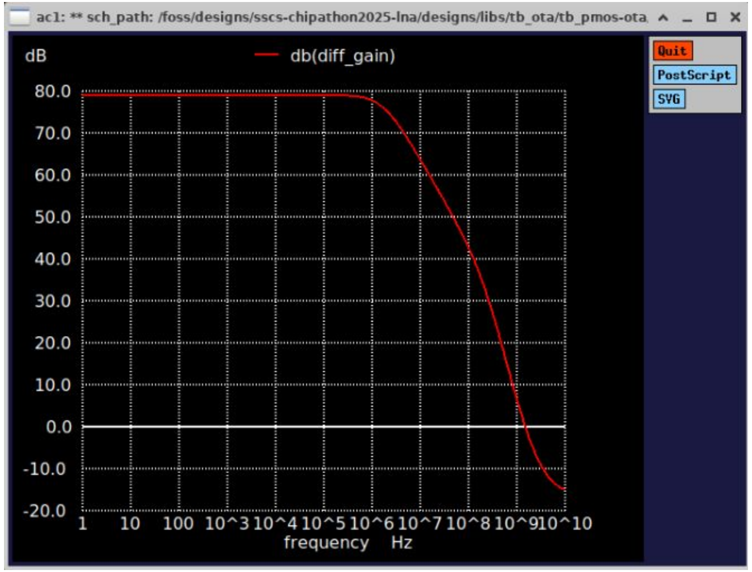




Power Consumption and PSRR

M6_Low Noise Anomaly

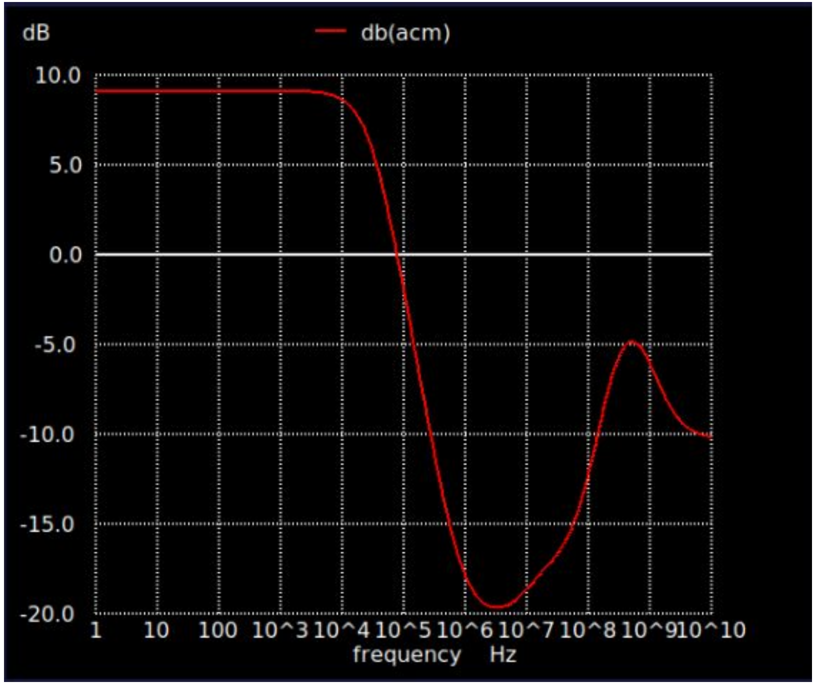




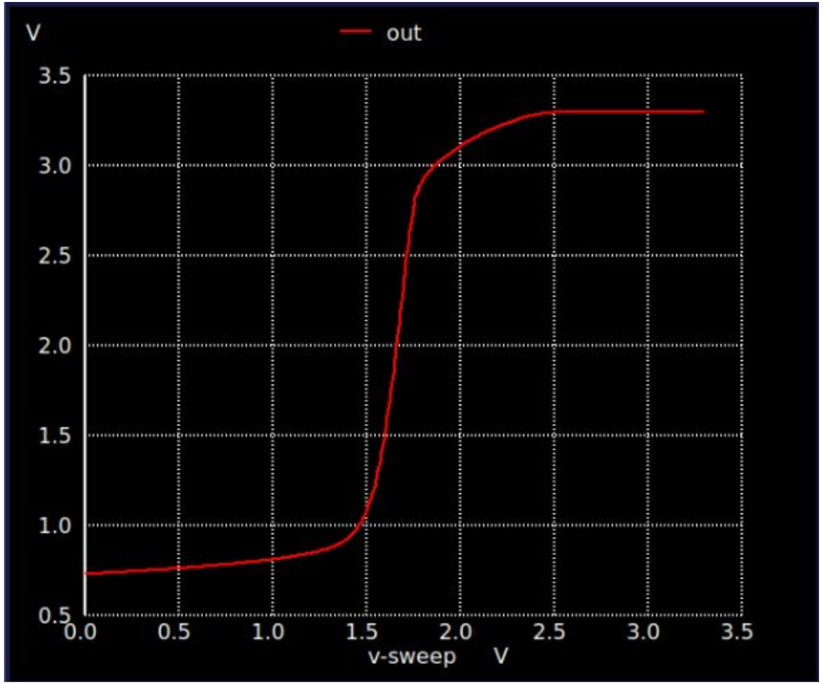
Operating Points	
Corner Case	Typical
Ibias	10uA
VDD	3.3 Volts

2-stage pmos OTA

M6_Low Noise Anomaly

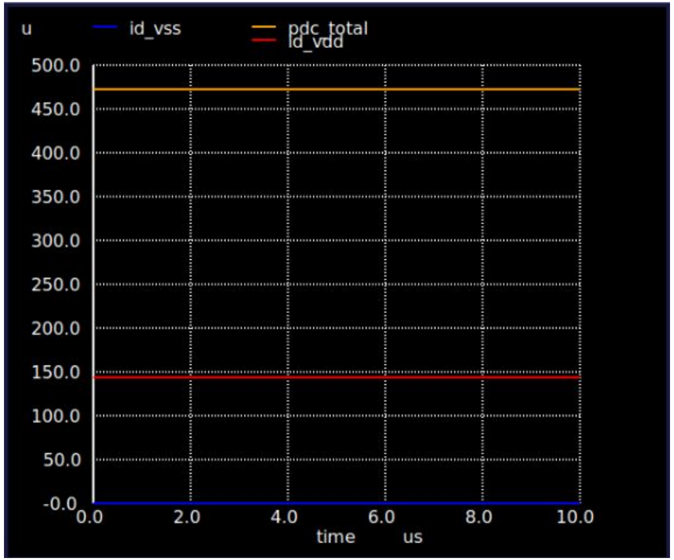


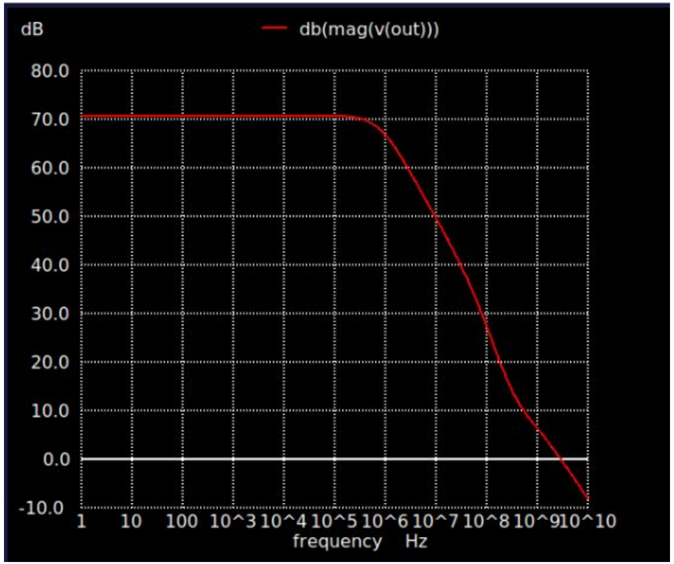
Common-Mode Gain



Input Common Mode Range

M6_Low Noise Anomaly

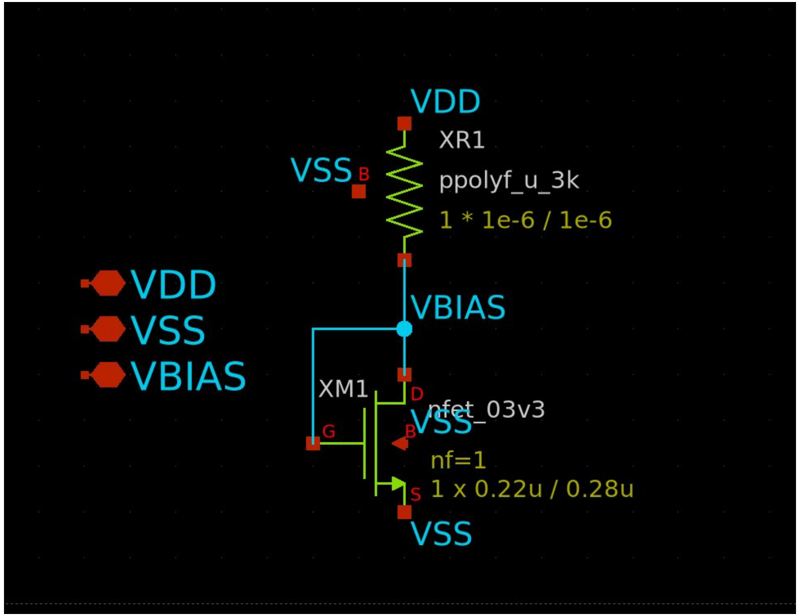


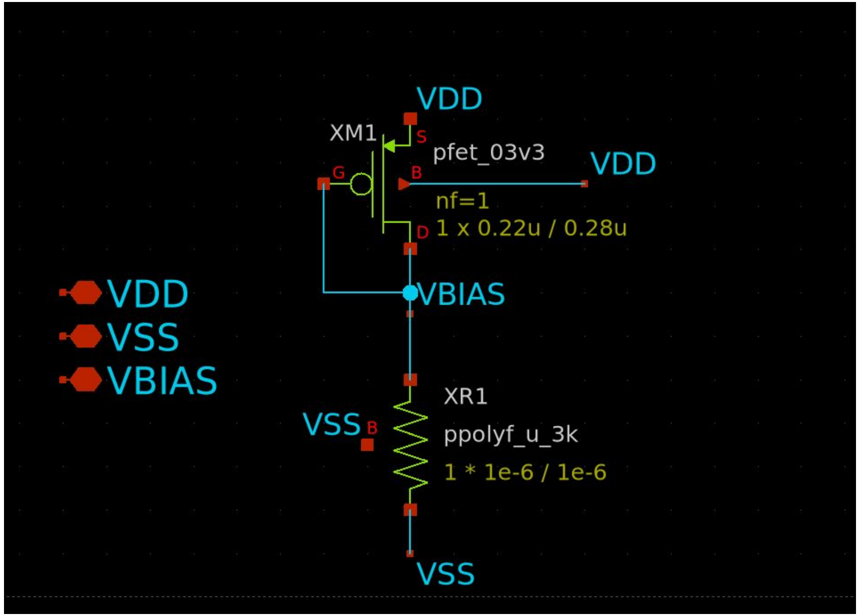


Power Consumption and PSRR

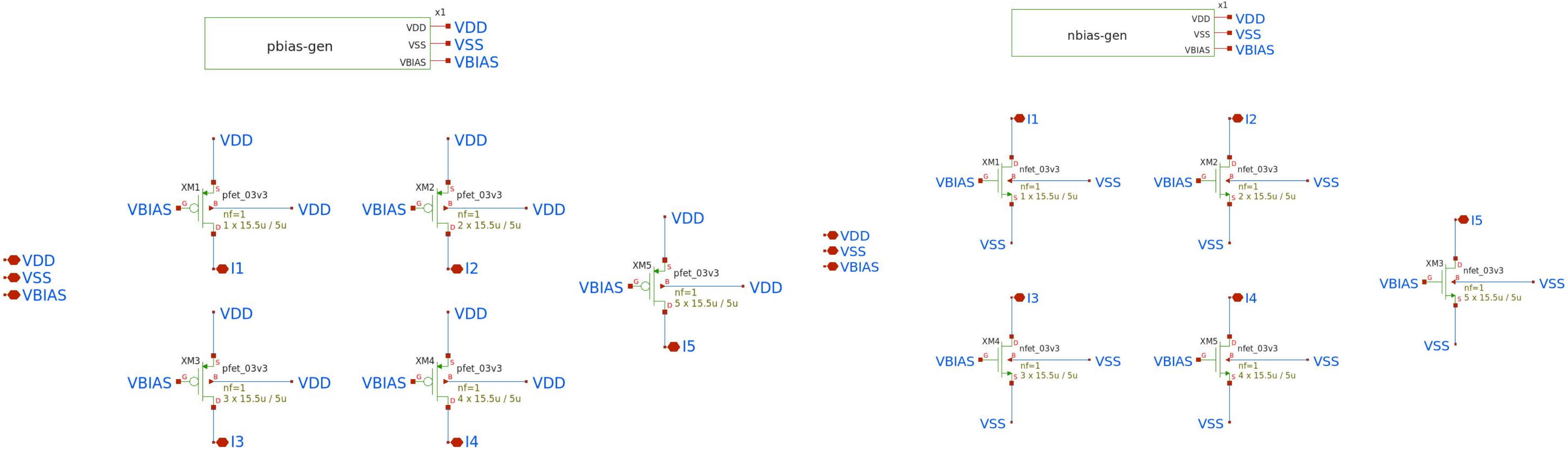
M6_Low Noise Anomaly

MOSFET Diode P and N

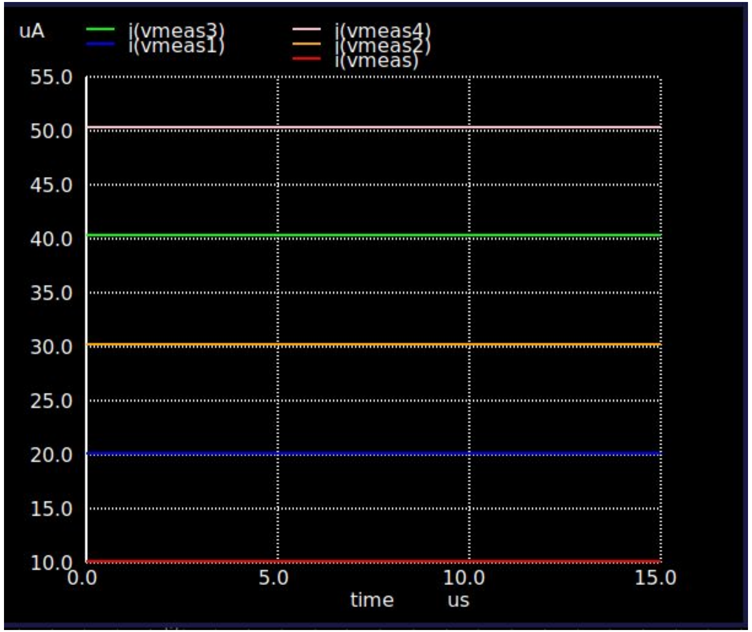




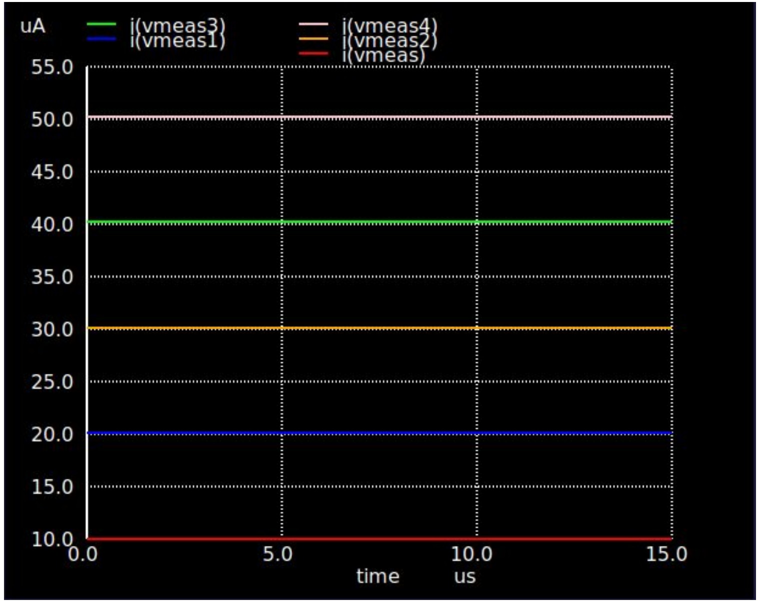
M6_Low Noise Anomaly - PMOS and NMOS Cascode Current Mirror



M6_Low Noise Anomaly - PMOS & NMOS Current Mirror Simulation



PFET Current Mirror



NFET Current Mirror